

Slide 1: Title Slide

- This is the cover page of the presentation.
 - Topic: Earthquake risk and safety situation of Punjab and Chandigarh.
 - Made by Anushka Thakur, a Master's student in Disaster Management.
 - This work was done as part of an internship at MGSIPA (a Punjab government institute), and submitted to Panjab University.
-

Slide 2: Introduction - Why Earthquakes Matter Here

Introduction

- Earthquakes happen when the Earth's big rock plates move and release energy, which shakes the ground.
 - The Indian Plate is slowly pushing into the Eurasian Plate (about 5 cm every year). This is also what created the Himalayas.
 - Because Punjab is near the Himalayas, and Chandigarh is even closer, both areas have a real earthquake risk.
 - Key numbers: 59% of India's land can face medium-to-strong earthquakes; Chandigarh is in 'Zone IV', which means high risk.
-

Slide 3: How an Earthquake Happens

Chapter 1: Fundamentals

- An earthquake happens in 4 simple steps: (1) Stress builds up in rocks, (2) the rock breaks (fault ruptures), (3) energy spreads out as waves, (4) the ground shakes.
 - Fault = a crack in the Earth's crust. Focus = the underground point where the break starts. Epicenter = the point on the surface right above the focus.
 - Earthquakes can be caused by moving tectonic plates, sudden slips in faults, volcanoes, big dams (reservoirs), or mining/explosions.
-

Slide 4: Seismic Waves - How Shaking Travels

Chapter 1: Fundamentals

- There are 4 types of earthquake waves: P-waves (fastest, least damage), S-waves (slower, moderate damage), Love waves and Rayleigh waves (slowest, but cause the most damage).

- Magnitude means how much energy was released - measured using the Moment Magnitude Scale (this has replaced the old Richter Scale).
- Intensity means how strong the shaking felt at a specific place - measured on a scale from I to XII.
- Special instruments called seismographs and accelerographs are used to record earthquake shaking.

Slide 5: Global Setting - Where Earthquakes Happen in the World

Chapter 2: Global Setting

- Earthquakes are not random - they mostly happen along the edges (boundaries) of tectonic plates.
- The 'Pacific Ring of Fire' has about 75% of the world's earthquakes.
- The 'Alpine-Himalayan Belt' (which includes northern India) is the second most earthquake-active area in the world.
- India's situation: the Indian Plate started colliding with the Eurasian Plate about 50 million years ago, and this collision is still going on today, still building the Himalayas.

Slide 6: India's Four Seismic Zones

Chapter 3: Seismic Framework of India

- India is divided into 4 earthquake risk zones by the government's BIS rules: Zone II (low risk) to Zone V (very high risk).
- Zone V (very high risk): North-East India, Jammu & Kashmir, Himachal Pradesh, Uttarakhand, North Bihar, Andaman & Nicobar.
- Zone IV (high risk): Chandigarh and northern parts of Punjab.
- Zone III (moderate risk): Central Punjab and most of the Indo-Gangetic Plain.
- Zone II (low risk): Southern and south-western districts of Punjab.

Slide 7: History of Major Earthquakes in India

Chapter 3: Seismic Framework of India

- This slide shows a list of India's biggest historical earthquakes, like 1897 Assam, 1905 Kangra, 1934 Bihar-Nepal, 1950 Assam-Tibet, 1993 Latur, 2001 Bhuj, and 2005 Kashmir.

- Not all big earthquakes happen at plate boundaries. Some, like Latur (1993), Jabalpur (1997), and Bhuj (2001), happened inside a normally 'stable' area, because old hidden faults woke up again.
 - Lesson for Punjab: even areas that are not directly on a plate boundary can still face serious earthquake risk.
-

Slide 8: Punjab - Close to the Belt, Not On It

Chapter 4: Punjab's Hazard Profile

- Punjab is located between a stable zone (Peninsular Shield) and the highly active Himalayas.
 - Punjab is not exactly on a plate boundary, but the stress from the Indian-Eurasian collision still reaches Punjab through the Indo-Gangetic Plain.
 - Punjab's ground is made of thick soft soil (called alluvium) deposited by rivers like Satluj, Beas, Ravi, and Ghaggar. This soft soil can make shaking feel stronger and can also cause 'liquefaction' (soil turning weak, almost like liquid, during a quake).
 - Main fault systems that affect Punjab include the Himalayan Frontal Thrust (HFT) and Main Boundary Thrust (MBT), which have the biggest influence.
-

Slide 9: District-wise Earthquake Zones of Punjab

Chapter 4: Punjab's Hazard Profile

- Punjab's districts are divided into three risk zones.
 - Zone IV (High risk): Pathankot, Gurdaspur, Hoshiarpur, Rupnagar, SBS Nagar, Mohali (SAS Nagar), and Chandigarh.
 - Zone III (Moderate risk): Amritsar, Tarn Taran, Kapurthala, Jalandhar, Ludhiana, Fatehgarh Sahib, Patiala, Sangrur, Barnala, Mansa.
 - Zone II (Low risk): Bathinda, Faridkot, Muktsar Sahib, Ferozepur, Fazilka.
 - In short: risk is highest in the north (near the Himalayas) and gets lower as you move south.
-

Slide 10: Punjab Has Felt Big Regional Earthquakes

Chapter 4: Punjab's Hazard Profile

- Punjab itself has never had a big earthquake starting inside it, but it has felt strong shaking from nearby big earthquakes many times (like Kangra 1905, Bhuj 2001, Kashmir 2005, Nepal 2015).
- Why the soil matters: Punjab's soft, thick soil and high groundwater level make shaking feel stronger, and raise the risk of liquefaction.
- This is why even earthquakes that happen far away (in the Himalayas) can still be felt strongly inside Punjab.

Slide 11: What Makes Punjab Vulnerable

Chapter 4: Punjab's Hazard Profile

- 'Vulnerability' means how much damage people, buildings, and infrastructure could suffer from an earthquake.
- Structural vulnerability: buildings made without engineering, old houses, schools, hospitals, bridges, flyovers, and industrial buildings that are not earthquake-safe.
- Non-structural vulnerability: too many people living close together, poor emergency planning, low public awareness, and possible failure of power, communication, water, and healthcare systems.
- Primary hazards (direct effects): ground shaking, cracks on the surface, buildings collapsing.
- Secondary hazards (effects after the main shake): liquefaction, ground sinking, fires, bridges breaking, damage to dams and canals.

Slide 12: Risk Across Punjab's Major Cities

Chapter 4: Punjab's Hazard Profile

- This slide compares earthquake risk across Punjab's cities using zones and colors.
- High risk (Zone IV): Chandigarh, Mohali, Rupnagar, Pathankot, Gurdaspur, Hoshiarpur.
- Moderate risk (Zone III): Amritsar, Jalandhar, Ludhiana, Patiala.
- Low-to-moderate risk (Zone II): Bathinda.
- Overall risk depends on the zone, how crowded the city is, how strong the buildings are, and the type of soil.

Slide 13: Two Earthquakes, Two Lessons

Chapter 6: Case Studies

- This slide compares two famous Indian earthquakes: 1905 Kangra earthquake and 2001 Bhuj earthquake.
 - 1905 Kangra: Magnitude 7.8, happened in Himachal Pradesh, killed over 20,000 people, caused by Himalayan plate collision.
 - 2001 Bhuj: Magnitude 7.7, happened in Gujarat, killed over 13,000 people, caused by an old hidden fault inside a normally stable area (intraplate).
-

Slide 14: What Both Earthquakes Teach Punjab

Chapter 6: Case Studies

- Kangra earthquake was felt strongly in Punjab and even cracked a structure (Ramgarhia Bunga) at the Golden Temple, Amritsar.
 - Bhuj earthquake proved that even 'safe' areas away from plate boundaries can face major earthquakes.
 - 4 lessons/actions suggested: enforce earthquake-safety building codes, check and fix old buildings, make important buildings (hospitals, schools) stronger, and improve public disaster preparedness.
 - After Bhuj, India created the NDMA (National Disaster Management Authority), a new disaster law (2005), the NDRF (National Disaster Response Force), and updated building safety codes.
-

Slide 15: Building Resilience Before the Next Quake

Chapter 5: Mitigation & Resilience

- Earthquakes cannot be stopped, but damage can be reduced through good planning and preparation.
 - Earthquake-resistant construction: buildings designed with balanced shapes, strong foundations, and flexible materials.
 - Retrofitting: making older buildings stronger using techniques like steel bracing and extra wall support.
 - BIS codes: official Indian building safety standards (like IS 1893, IS 4326, IS 13920) that guide safe construction.
 - Other steps: mapping risk areas using GIS technology, community awareness and mock drills, and better teamwork between government disaster agencies (NDMA, PSDMA, district offices).
-

Slide 16: Findings & Recommendations

Chapter 7: Findings & Recommendations

- Key findings: Punjab is not exactly on a plate boundary but is close enough to face real risk; it falls in Zones II, III and IV; its soft soil increases shaking and liquefaction risk; fast city growth and unsafe buildings are increasing the danger; monitoring by agencies like NCS, IMD, NDMA and PSDMA has improved, but gaps remain.
 - Key recommendations: strictly follow earthquake-safety building codes for new construction; check and strengthen schools, hospitals, bridges and old heritage buildings; make and update district-level risk maps; limit building in high-risk fault areas; add more earthquake-monitoring stations, especially in north Punjab; run regular public awareness programs and mock drills; and improve coordination between NDMA, PSDMA and district authorities.
-

Slide 17: Conclusion - A Resilient Punjab is a Planned Punjab

Conclusion

- Even though Punjab is not directly on a plate boundary, its closeness to the Himalayas and its soil type still make it vulnerable to earthquakes.
 - The 1905 Kangra and 2001 Bhuj earthquakes show both the danger and how good preparation can help.
 - Building real safety means combining: safety codes, retrofitting old buildings, hazard mapping, better monitoring, and public awareness - all working together.
 - Main message: If disaster risk reduction is added into today's development planning, Punjab will be safer in the future.
-

Slide 18: Internship Experience at MGSIPA, Chandigarh

Chapter 8: Internship Experience

- This report was made during the author's internship at MGSIPA (a government training institute in Chandigarh).
- The internship helped in preparing an Institutional Disaster Management Plan (IDMP) for the institute.
- Work included: hazard identification and risk assessment, making hazard maps and tables, preparing emergency response plans, and helping with a Pre-Monsoon Preparedness Workshop.
- Real example mentioned: the 2005 Kashmir earthquake's shaking cracked walls at a hall in Chandigarh's Sector 42, showing that Chandigarh's high-risk (Zone IV) status is real, not just theoretical.

Slide 19: Thank You / Closing Slide

- This is the final 'Thank You' slide.
 - It repeats the topic and the author's name and background.
 - It also lists the official sources used for this research: BIS, NDMA, NCS, IMD, and PSDMA.
-